Companion LXXXI: Diffusion–Optimal Transport Unification and Neural Transport Maps for Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Diffusion models and optimal transport (OT) provide two complementary perspectives on probability distributions: diffusion models describe generative stochastic dynamics, while OT characterizes deterministic transport maps. Recent advances have shown that these two viewpoints can be unified through stochastic control, Schrödinger bridges, and score-based transport geometry. Building on the neural Schrödinger flows developed in Companion LXXX, this work introduces a unified diffusion–OT framework for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, inducing characteristic changes in both the deterministic OT map and the stochastic diffusion flow. The resulting neural transport maps provide a powerful tool for modeling RDCC signatures, enabling likelihood-free inference, generative modeling, and transport-based parameter constraints from weak-lensing topology.

1 Introduction

Persistence diagrams encode the topological structure of weak-lensing convergence fields. Previous Companions introduced Wasserstein geometry, entropic optimal transport, Schrödinger bridges, score-based diffusion models, and neural Schrödinger flows as tools for analyzing the Relaxation-Driven Cyclic Cosmology (RDCC).

Diffusion models provide stochastic generative dynamics, while optimal transport provides deterministic transport maps. Unifying these perspectives yields a powerful framework in which:

- diffusion models approximate OT maps in the zero-noise limit,
- OT maps arise as the mean flow of Schrödinger bridges,
- score fields encode the gradient of the log-density,
- neural SDEs learn both stochastic and deterministic transport geometry.

In RDCC, the relaxation mechanism modifies both the deterministic and stochastic components of transport, producing characteristic signatures in the unified diffusion–OT geometry.

2 Optimal Transport Maps for Persistence Diagrams

Let ρ_0 and ρ_1 be distributions of persistence diagrams. The OT map T solves

$$T = \arg\min_T \int \|D - T(D)\|^2 \, d\rho_0(D), \tag{1}$$

subject to $T_\# \rho_0 = \rho_1$.

For persistence diagrams, T acts on birth-death coordinates and encodes the deterministic geometric deformation between topological ensembles.

3 Diffusion Models as Stochastic Transport Maps

A diffusion model defines the reverse-time SDE

$$dD_t = -\beta(t) s(D_t, t) \, dt + \sqrt{\beta(t)} \, d\bar{B}_t. \tag{2}$$

The mean flow of this SDE approximates the OT map:

$$\mathbb{E}[D_t] \approx T(D_0). \tag{3}$$

Thus diffusion models provide a stochastic relaxation of OT maps.

4 Unified Diffusion–OT Geometry

The unified framework is governed by the controlled SDE

$$dD_t = [-\varepsilon(t) s(D_t, t) + u(D_t, t)] \, dt + \sqrt{2\varepsilon(t)} \, dB_t, \tag{4}$$

where

- $s(D_t, t)$ is the score field,
- $u(D_t, t)$ is a deterministic OT-like control,
- $\varepsilon(t)$ interpolates between OT ($\varepsilon \to 0$) and diffusion.

This yields a continuum between:

$$\text{deterministic OT} \quad \leftrightarrow \quad \text{stochastic diffusion.}$$

5 RDCC Modification of Unified Transport Geometry

In RDCC, the convergence evolves as

$$\kappa_{\mathrm{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\mathrm{rel}}} \right)^\alpha \right]. \tag{5}$$

This modifies:

- the score field $s(D, t)$,
- the deterministic OT control $u(D, t)$,
- the diffusion schedule $\varepsilon(t)$,
- the curvature of transport trajectories,

- the barycentric flow across redshift.

RDCC induces:

- enhanced deterministic transport at large scales,
- suppressed stochasticity at small scales,
- a characteristic turnover near ℓ_{rel},
- scale-dependent deformation of transport maps.

6 Conclusion

Diffusion models and optimal transport provide complementary perspectives on the geometry of probability distributions. Neural Schrödinger flows unify these perspectives into a single framework for learning stochastic and deterministic transport processes. In the Relaxation-Driven Cyclic Cosmology, the relaxation mechanism modifies both components of transport in a scale-dependent manner, producing distinctive signatures in weak-lensing topology. The present Companion establishes the foundation for transport-based RDCC parameter inference, generative modeling, and unified geometric analysis of topological data.

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